



SukaSeafood

SukaSeafood — Iteration 1 Build Plan

Mentor Review — 24 August 2026

1. Iteration 1 Goal & Frozen Scope

Iteration 1 objective

Prove the core SukaSeafood decision-support journey for five supported Malaysian seafood species, allowing users to identify or search seafood, understand its sustainability, contextualise price and supply, and receive sustainable recommendations based on cooking method.

The target journey is:

Search / Identify → Confirm Seafood → Understand Sustainability → Contextualise Price & Supply → Match Cooking Method → Choose

The four Iteration 1 epics are:

Epic 1 — Seafood Discovery & Identification

User question: *What fish is this?*

Two entry points are provided:

Search

- User types a seafood name.

- This allows SukaSeafood to be used **before leaving home or while travelling to the grocery store.**
- Search terms map to one canonical seafood record.

Computer Vision

- User photographs seafood at the point of purchase.
- Model returns the predicted species, confidence and top alternatives.
- User confirms the identification before downstream information is attached.

Epic 2 — Sustainability Intelligence

User question: *Is it sustainable, and why?*

- WWF Save Our Seafood classification
- Origin
- Production method
- Relevant explanation/notes
- Deterministic source-backed classification rather than a SukaSeafood-created sustainability score

Epic 3 — Price & Supply Intelligence

User question: *Is this a reasonable choice for me?*

- Latest observed price
- Recent price context
- Local/location context
- Broader marine fish landing/supply context

Iteration 1 does **not** include price forecasting or species-level supply prediction.

Epic 4 — Cooking Intent

User question: *What should I buy for what I'm cooking?*

- User selects a cooking method such as grilling, soup, curry, rice dish or stir-fry.
- SukaSeafood recommends suitable seafood.
- Sustainability remains a core ranking/input to the recommendation.
- Natural-language cooking intent/NLP is outside Iteration 1.

The presentation deck identifies these four as the Iteration 1 “must-have epics” and presents them as one complete user journey.

Supporting platform functions

These support the journey but are **not additional epics**:

- Account/authentication
- Favourites

- History
- Explicit purchase confirmation
- Sustainable Choice Journey
- Saved default location
- GPS/manual location fallback

These functions are included in the current implementation scope in the team build plan.

Supported species

The initial scope is deliberately limited to:

Kembung/Pelaling · Bawal Hitam · Ikan Merah · Tilapia · Kerapu Bintik

2. Core Data Architecture

All four epics use a **single canonical seafood record**.

Canonical seafood entity

Each supported species receives a stable:

`seafood_item_id / fish_id`

All data sources map to this identifier.

For example:

```
1 fish_id
2 scientific_name
3 primary_common_name
4 Malaysian/local names
5 aliases
6 CV class
7 WWF record
8 PriceCatcher mapping
9 supply context
10 cooking suitability
```

This allows the system to connect:

Search / CV → Sustainability → Price & Supply → Cooking

without creating separate records for the same fish.

3. Build Order, Dependencies & Definition of Done

Step	Build	Dependency / Parallel Work	Definition of Done
1	Database + canonical seafood mapping	No dependency on final CV model	Database deployed; five stable seafood IDs; aliases and source mappings loaded
2	Core API + seed data	Starts once DB baseline exists	Search, seafood profile, location and source/version endpoints work
3	WWF + Price + Supply + Cooking data	Can run in parallel once IDs exist	WWF lookup, price/supply context and reviewed cooking suitability are available through APIs
4	Search experience	Uses canonical seafood IDs	User can type a supported fish name and reach its profile
5	Frontend core journey	Can use mock CV response	Search → sustainability → price/supply → cooking works in the UI
6	CV model + / <code>identify</code> adapter	CV develops in parallel with frontend/backend	Top-3 prediction + confidence + model version returned; user confirms result
7	Supporting account functions	Core journey stable	Favourite, history, purchase confirmation, journey and default location work
8	Integration + QA	All components available	End-to-end journey, permissions and fallback states pass
9	Demo hardening	Only after acceptance criteria pass	Full I1 demonstrated without unsupported claims or hidden fallback behaviour

This preserves the original build plan's dependency logic while making the **canonical mapping, search and four-epic journey explicit**.

4. Key Milestones & Dates

Monday 24 August — Build Start

24 Aug — Foundation Kick-off

Priority: Database and data contracts

Complete:

- Five canonical seafood IDs
- Seafood master table
- Alias/local-name mapping
- Dataset-to-`fish_id` mapping design
- API contracts agreed
- Frontend/CV interface contract agreed

Parallel work begins:

- WWF data curation
 - Price/lookup integration
 - Cooking suitability curation
 - CV model preparation
-

Tuesday 25 August

25 Aug — Core Data + API Baseline

Complete:

- Database deployed
- Seed data loaded
- Search endpoint
- Seafood profile endpoint
- Location endpoint
- Source/version metadata
- Mock `/identify` contract

Milestone: A typed search term can return one canonical seafood record.

Wednesday 26 August

26 Aug — Core Decision Data

Complete:

- WWF structured dataset integrated
- PriceCatcher integrated
- Item/Premise lookup integration
- Fish landings/supply context integrated
- Cooking suitability table integrated

Milestone: A manually selected seafood item can show the four core data layers.

Thursday 27 August

27 Aug — Frontend Core Journey

Complete:

Search → Seafood profile → Sustainability → Price/Supply → Cooking Method → Recommendation

The frontend should work with mocked CV results.

Milestone: Full Iteration 1 journey works without waiting for the final CV model.

Friday 28 August

28 Aug — CV + Account Integration

Complete:

- CV model baseline
- `/identify` adapter
- Top-3 predictions
- Confidence score
- User confirmation
- Favourite
- History
- Purchase confirmation
- Saved location

Milestone: Search and camera become two entry points into the same seafood record.

Saturday–Sunday 29–30 August

29–30 Aug — Integration + QA

Focus on:

- API/frontend integration
- Search edge cases
- CV edge cases
- WWF ambiguity
- Sparse price data
- Supply-data limitations
- Unverified cooking records
- GPS/manual location fallback
- Guest vs registered-user permissions
- Five supported species

No new features should be introduced during this period.

5. Pre-Deployment Readiness

Deadline: Monday 31 August 2026 — 11:55pm MYT

By this deadline, Iteration 1 should be **deployment-ready**, not merely partially implemented.

Pre-deployment readiness criteria

Product

- Four I1 epics connected
- Search functional
- CV functional or stable mock fallback
- Five supported species covered
- Core user journey demonstrable

Data

- Canonical IDs stable
- WWF records verified
- Price data verified
- Supply context validated
- Cooking suitability reviewed

Technical

- APIs stable
- Frontend integrated
- Authentication/permissions working where applicable
- Environment/configuration ready for deployment

QA

- Critical happy path passes
- Major fallback states tested
- No unsupported data claims displayed
- No known blocking defects

Pre-deployment readiness means the system is safe to deploy and test as an integrated I1 product.

6. Iteration Build Deadline

Thursday 3 September 2026 — 11:55pm MYT

The additional time between readiness and final iteration submission is for:

- fixing defects identified during readiness
- improving stability
- completing remaining non-blocking integration work
- final QA
- final demo validation
- documentation and evidence
- confirming that the Definition of Done is met

31 August = deployment-ready.

3 September = Iteration 1 build complete and submitted.

7. Ownership

Workstream	Accountable Owner	Immediate Deliverable
Frontend	To confirm	Core screens + search + mock-CV flow connected to API
Backend / Data	To confirm	DB + canonical IDs + APIs + WWF/Price/Supply logic
CV	To confirm	Five-class model + validation + <code>/identify</code>
WWF / Cooking Data	To confirm	Verified WWF extraction + cooking suitability dataset
QA / Technical Integration	To confirm	Integration test matrix + end-to-end sign-off
Presentation / Documentation	To confirm	Current documentation + demo evidence

One person should be accountable for each workstream, with a support/reviewer where needed. This matches the ownership structure already proposed in the team build plan.

8. Dependencies & Risks

Computer vision not ready

Mitigation: Frontend/backend continue using mock CV responses until the real model is available.

Search and CV produce different names

Mitigation: Both must resolve to the same canonical `fish_id`.

WWF source ambiguity

Mitigation: Manual verification and an explicit **Undetermined** state rather than guessing.

Sparse price data

Mitigation: Return **Insufficient** / **Unavailable** rather than inventing or extrapolating a price.

Supply data limitations

Mitigation: Use OpenDOSM Fish Landings as **broader supply context**, not unsupported species-level forecasting.

Cooking data not verified

Mitigation: Do not expose the recommendation until the suitability mapping has been reviewed.

GPS unavailable

Mitigation: Manual location selection.

CV model late

Mitigation: Mock CV remains a valid development fallback.

The original build plan already identifies the principle that unavailable, insufficient and undetermined states are legitimate product outcomes rather than states to hide.

9. Definition of Done

Product acceptance

Iteration 1 is complete when a user can:

1. Search

Type a supported seafood name and reach the correct canonical seafood record.

2. Identify

Alternatively scan a supported fish and receive:

- predicted species
- confidence
- top alternatives
- user confirmation

3. Understand

View:

- WWF sustainability classification
- origin
- production method
- explanation

4. Contextualise

View:

- observed price
- recent/local price context
- broader supply context

5. Match

Select a cooking method and receive:

- suitable seafood
- sustainability information
- recommendation rationale

6. Continue the journey

Save/favourite, record history or confirm a purchase where the supporting account functions are enabled.

Technical acceptance

- All four epics use the same canonical seafood IDs.
 - Five supported species work end-to-end.
 - Search and CV both resolve into the same backend model.
 - External source data is traceable.
 - Fallback states work.
 - APIs and frontend are integrated.
 - Critical end-to-end tests pass.
 - No unsupported claims are presented as facts.
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10. What Is Explicitly Outside Iteration 1

To protect the scope, the following remain future work:

- **Smart Swap**
- advanced seafood comparison
- natural-language cooking intent / NLP
- price forecasting
- species-level supply prediction
- advanced marine biodiversity/context modelling
- personalised recommendations
- basket optimisation
- Demand-Side Impact Simulator
- Sustainable Choice Journey gamification beyond the supporting I1 functionality
- expansion beyond the five supported species

This keeps the I1 build consistent with the frozen scope in the team's current build plan.

